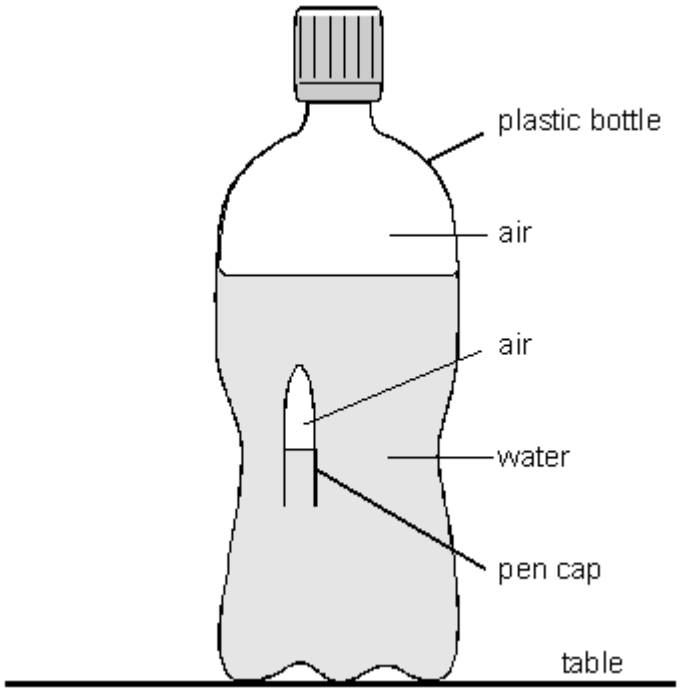
Q1.

A pen cap floats in a plastic lemonade bottle three-quarters full of water.
If you squeeze the bottle the pen cap sinks to the bottom.
If you then let go of the bottle, the pen cap floats to the surface.



(a) When the bottle is squeezed what, if anything, happens to:

(i) the distance between the air molecules inside the bottle?

.....

1 mark

(ii) the distance between the water molecules inside the bottle?

.....

1 mark

(iii) the pressure of the air trapped inside the pen cap?

.....

1 mark

(iv) the volume of the air trapped inside the pen cap?

.....

1 mark

(b) Explain why the pen cap sinks when you squeeze the bottle.

.....

.....

.....

.....

2 marks
Maximum 6 marks

Q1.

(a) (i) they get closer **or** it gets less

1

(ii) nothing **or** same distance

1

(iii) it increases

1

(iv) it decreases

1

(b) water flows into the cap

*accept 'water flows or is pushed **or** got into the cap'*
***or** 'the air in the cap takes up less space'*
accept 'the air in the cap is under pressure'

1

any **one** from

- increasing the density
- less upthrust
- pen cap now less buoyant

accept 'increasing the weight'
*do **not** accept 'the pen cap gets heavier'*

1

[6]